

Claim dependencies

Repository source: proof/CLAIM_DEPENDENCIES.md

Purpose

This document exposes the logical dependency structure of the proposed theorem. It separates sourced classical facts, human mathematical reductions, and finite machine-checked claims.

Status labels

- **Sourced:** imported from an identified mathematical reference.
- **Paper proof:** derived in `paper/contact_geometric_bound.tex`.
- **Exact certificate:** checked with integer and rational arithmetic.
- **Independent audit:** checked by a separately organized numerical path.
- **Release infrastructure:** checks consistency, provenance, or presentation.
- **External review needed:** no independent subject-matter review has been recorded.

Dependency graph

```
C0 constant-width identities and scaling
|
+--> C1 minimum-circumsphere contact balance and Jung range
    |
    +--> L0 lower-branch radial transport and tangent-gap reduction
        |
        |   +--> L1 exact rational propagation cells and row partition
        |   |
        |   |   +--> L2 unique bottleneck 03B02
        |   |
        |   +--> N0 four-contact near-Jung deficiency budget
        |   |
        |   |   +--> N1 affine tetrahedral coordinates
        |   |   |
        |   |   |   +--> N2 determinant transfer
        |   |   |   +--> N3 dominant-edge coordinate-width lemma
        |   |   |
        |   |   +--> N4 exact forced fan
        |   |   |
        |   |   |   +--> N5 physical forced-core lower bound
        |   |   |
        |   |   +--> N6 three antipodal cap-pair bounds
        |   |   |
        |   |   |   +--> N7 exact six-cap disjointness
        |   |   |   |
        |   |   |   |   +--> N8 near-Jung lower bound and strict handoff
        |   |   |
        |   |   +--> L2 + N8 + gap-free partition --> T0 universal coefficient
```

Claim ledger by dependency

ID	Claim	Evidence	Verification class	Depends on
C0	Width-one support identity, difference body, diameter one, cubic scaling	Paper Section 2; classical sources	Sourced; paper proof	-
C1	Minimum-ball contacts balance; $1/2 \leq R \leq \sqrt{6}/4$; four contacts for $R^2 > 1/3$	Paper Section 2	Sourced; paper proof	C0
C2	Opposite-point identity $x - n \in K$ and support inequality	Paper Lemma 2.1	Paper proof	C0
L0	Divergence transport inequality and tangent majorant	Paper Section 3	Paper proof	C0, C2
L1	Rational propagation, exact cap rings, and complete lower row table	Four NPZ cells, reference handoff JSON, exact verifier	Exact certificate	L0
L2	03B02 is the unique lower-branch bottleneck at R_*	Final JSON and byte-identical row reconstruction	Exact certificate	L1
N0	Four-contact deficiency budget $S \leq (3 - 8R^2)/(4R^2 - 1)$	Paper Section 4	Paper proof	C1
N1	Affine tetrahedral reference coordinates and explicit Gram formulas	Paper Section 4	Paper proof	N0
N2	$\det H \geq 1 - S/2 - S^2/2$	Paper Proposition 4.1; exact assembly inputs	Paper proof; exact evaluation	N1
N3	Dominant-edge diagonal bounds and three coordinate-width lower bounds	Paper Section 5; certificate JSON	Paper proof; exact evaluation	N0, N1
N4	98,306-ray fan satisfies every exact margin and inherited multiplier condition	Fan NPZ files, orbit audit TSV, exact verifier	Exact certificate	N0
N5	Determinant-transferred forced-core volume lower bound	Fan volume and determinant fields	Exact certificate	N2, N4
N6	Three optimized antipodal cap-pair lower bounds	Section hull TSV, cap fields, exact verifier	Exact certificate	N3, N4
N7	Interiors of all six caps are pairwise disjoint	Cross-axis disjointness fields and exact verifier	Paper proof; exact certificate	N3, N6

ID	Claim	Evidence	Verification class	Depends on
N8	Near-Jung branch exceeds the theorem upper enclosure by a strict positive margin	Final JSON and exact branch assembly	Exact certificate	N5, N6, N7
T0	Universal coefficient equals the 03B02 coefficient	Gap-free split and exact branch comparison	Exact certificate plus paper reduction	L2, N8
A0	Binary64 Python agrees with principal exact values	<code>audit_universal_bound.py</code> and transcript	Independent audit	L1, N4-N8, T0
A1	Base-R implementation agrees with principal values	<code>audit_universal_bound.R</code> and transcript	Independent audit	L1, N4-N8, T0
A2	Seven corrupted objects are rejected	Mutation suite and transcript	Adversarial audit	L1, N4, N6-N8, T0
D0	Paper theorem constants and lower table match the JSON	<code>paper/reconcile_certificate.py</code>	Release infrastructure	L1, T0
D1	Exact-constants appendix is generated from JSON	<code>proof/generate_exact_constants.py</code>	Release infrastructure	L1, N4-N8, T0
E0	Full theorem independently reviewed	No record yet	External review needed	C0-T0

Universal theorem assembly

The theorem uses the closed partition

$$\left[\frac{1}{2}, R_*\right] \cup \left[R_*, \frac{\sqrt{6}}{4}\right].$$

The lower branch proves the 03B02 coefficient throughout the first interval. The near-Jung branch proves a strictly larger coefficient throughout the second interval. The common endpoint causes no gap because both branch conditions are valid there.

The finite certificate checks the exact comparison. The manuscript must still justify that every width-one body has a circumradius in the Jung interval and falls under the corresponding branch reduction.

Audit dependencies are not proof dependencies

The numerical audits, PDF checks, and continuous-integration jobs do not feed into the logical proof of T0. They are error-detection and release-integrity layers. A failed audit blocks release because it indicates a discrepancy, but a passing audit does not substitute for the exact path or the paper proof.

Highest-value external review points

The most consequential points for independent mathematical review are:

1. passage from arbitrary bodies to the regularized radial inequality;
2. completeness and direction of each lower-branch propagation implication;
3. the deficiency budget and determinant transfer;

4. the dominant-edge coordinate-width lemma;
5. the geometric interpretation of the fan and section hulls;
6. the cap-volume formula and pairwise-disjointness argument; and
7. the final universal quantifier across the branch split.